

Graph-Guided RAG for Automatic Cluster Labeling*

Course Project: Mathematical Models in Advanced Information Retrieval

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Abstract

Automatic cluster labeling using large language models (LLMs) produces readable descriptions but often suffers from genericness and a lack of grounding. Furthermore, feeding an entire cluster into an LLM’s context window is noisy and computationally expensive. This project addresses the challenge of selecting the optimal subset of representative documents to prompt an LLM, thereby minimizing hallucinations and maximizing label groundedness. We propose a Graph-Guided Retrieval-Augmented Generation (RAG) framework that uses graph theory to filter outliers. Our pipeline constructs a graph where nodes are documents and edges represent semantic similarity, and applies PageRank to identify the core documents that are semantically central to the cluster. We then use these top- k central documents as the specific context for an LLM to generate a descriptive label. To ensure robustness, the framework is evaluated on four diverse text classification datasets (BBC News, Banking77, 20 Newsgroups, and AG News) using Groq’s LLaMA-3.1-8b model. We assess the generated labels against ground-truth categories using semantic similarity and lexical token-overlap F1 across a comprehensive hyperparameter sweep.

*Project code and implementation details are available at: https://github.com/maya-1807/Automatic_Cluster_Labeling

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1 Introduction

The automatic labeling of clusters is a fundamental task in information retrieval. Recently, large language models (LLMs) have been increasingly utilized for this task because they produce highly readable descriptions. However, this approach often suffers from genericness and a lack of grounding. Instead of reflecting the specific nuances of the documents within a cluster, the generated labels frequently rely heavily on the model’s pre-existing priors [1].

A significant technical bottleneck exacerbates this issue: feeding an entire cluster of documents into an LLM’s context window is not only computationally expensive but also introduces substantial noise. When models are provided with noisy or unrepresentative context, they become highly prone to hallucinations.

To overcome these limitations, this project introduces a Graph-Guided Retrieval-Augmented Generation (RAG) framework. We address the challenge of selecting the optimal subset of representative documents to prompt an LLM. By constructing a semantic similarity graph for each cluster and applying graph centrality metrics (specifically PageRank), our pipeline identifies the core documents that best represent the cluster’s overall theme. Using only these top central documents as context for the LLM minimizes hallucinations and maximizes the groundedness of the resulting labels.

2 Related Work

Statistical Methods Statistical Methods Early approaches to the automatic labeling of clusters relied on statistical tests to extract distinguishing terms directly from the text. [2]. While these traditional methods guarantee that the labels are grounded in the actual documents, the resulting concatenated terms often lack grammatical coherence and readability. Furthermore, they are inherently limited by the independence assumption of the Bag-of-Words model, often failing to capture latent thematic structures.

Graph-Based Ranking Graph-Based Ranking Researchers have demonstrated that structural re-ranking can effectively identify authoritative documents within a corpus. For example, Kurland and Lee (2010) utilized graph centrality measures, such as PageRank, to re-order documents based on their semantic influence. [3]. While their original framework induced links using language models, our approach adapts this intuition by utilizing dense sentence embeddings to capture deeper semantic relationships in the graph construction.

Generative Labeling Generative Labeling With the advent of Large Language Models (LLMs), recent studies have shifted towards generative methods for cluster labeling. [4]. These models produce highly fluent and human-readable descriptions. However, while fluent, these generative models are prone to hallucinations when provided with noisy or unrepresentative context. This necessitates a retrieval-augmented generation (RAG) framework to perform hard-filtering of outliers, ensuring the model remains grounded in the cluster’s specific nuances rather than its pre-training priors.

3 Proposed Methodology

Our proposed Graph-Guided Retrieval-Augmented Generation (RAG) framework consists of a multi-stage pipeline designed to filter out noisy documents and identify the most representative texts for cluster labeling. The pipeline operates in three main stages: Document Embedding and Graph Construction, Centrality-Based Selection, and Generative Labeling.

3.1 Embeddings and Graph Construction

In the first stage, we represent the textual content of the documents as dense vectors. We embed the documents using the `all-MiniLM-L6-v2` model from the `sentence-transformers` library. Once the embeddings are generated, we construct a similarity graph for each individual cluster. In this graph, the nodes represent the documents within the cluster, and the edges represent semantic similarity. We draw an edge between two nodes if the cosine similarity between their respective embeddings exceeds a predefined similarity threshold.

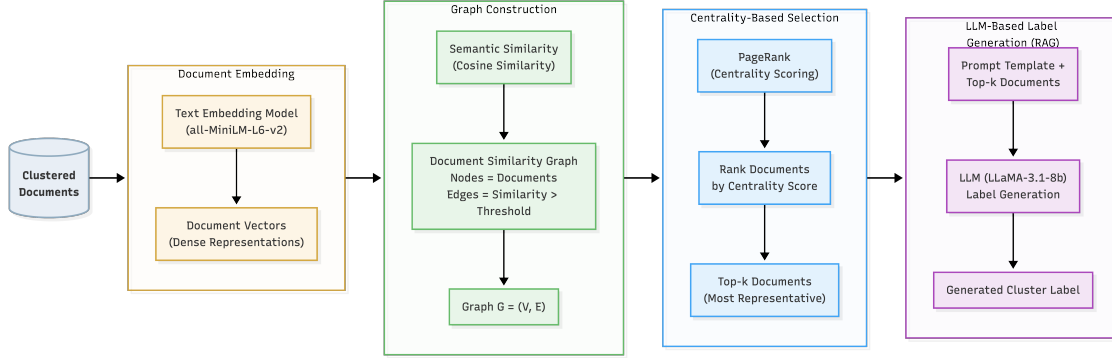


Figure 1: Overview of the proposed pipeline

3.2 Centrality-Based Selection

After constructing the similarity graph, we apply graph theory to filter outliers and identify the core documents. We run a weighted PageRank algorithm on the cluster’s graph to calculate a centrality score for each node. Documents with high PageRank scores are considered semantically central to the cluster, as they share high similarity with many other documents in the same group. We then select the top- k highest-scoring documents to serve as the representative subset for the cluster.

3.3 Generative Labeling via RAG

In the final stage, we utilize the selected top- k central documents as the specific context for a Large Language Model (LLM). For this task, we prompt the Groq `llama-3.1-8b-instant` model. By restricting the LLM’s context to only the most central documents (with a maximum limit of 300 words per document), we significantly reduce the noise and computational expense associated with processing the entire cluster. This focused context guides the LLM to generate a concise, highly grounded descriptive label for the cluster.

4 Experimental Setup

4.1 Datasets

To ensure the robustness of our framework across different text types, document lengths, and cluster granularities, we evaluated the pipeline on four diverse text classification datasets.

- **BBC News [5]:** A collection of 2,225 full news articles published between 2004 and 2005. The dataset is divided into 5 broad topic categories: business, entertainment, politics, sport, and tech.
- **Banking77 [6]:** A fine-grained intent detection dataset consisting of 13,083 short customer service queries from the banking domain. The documents are brief user messages categorized into 77 specific intent clusters (e.g., card arrival, lost or stolen card, and transfer fee charged).
- **20 Newsgroups [7]:** A classic text classification benchmark containing approximately 18,000 posts from 20 different Usenet newsgroups. Each document represents the body of a message, spanning topics such as computer science, recreation, politics, and religion.
- **AG News [8]:** A large-scale news classification dataset comprising approximately 120,000 articles. The documents are very short news summaries (typically 1-2 sentences) classified into 4 broad categories: World, Sports, Business, and Sci/Tech.

4.2 Evaluation Metrics

To quantitatively assess the quality of the generated cluster labels, we compared them against the ground-truth categories using two complementary metrics:

- **Semantic Similarity:** This metric captures the conceptual alignment between the generated label and the ground truth. We computed the cosine similarity between the dense vector embeddings of both labels, encoding the text with a `sentence-transformers` model. This allows us to reward generated labels that capture the correct underlying meaning, even if they use a different vocabulary than the ground truth.
- **Token Overlap F1:** To measure strict lexical agreement, we calculated the F1 score between the tokenized versions of the generated and ground-truth labels. By treating the labels as bags of words and computing the precision and recall of the overlapping tokens, this metric provides a concrete measure of literal, word-for-word accuracy.

4.3 Implementation and Hyperparameter Sweep

To rigorously evaluate our pipeline and prevent data leakage, we implemented a robust hyperparameter tuning process using an 80/20 stratified split for development (dev) and testing (test). The documents in the dev split were embedded once and cached to optimize computational efficiency during the sweep.

We conducted a comprehensive grid search over three key hyperparameters on the dev split:

- **Similarity Threshold:** {0.3, 0.5, 0.7}
- **Top- k Central Documents:** {3, 5, 10}
- **PageRank Damping Factor (α):** {0.5, 0.85, 0.95}

For each dataset, we evaluated all combinations of these hyperparameters on the dev split. The optimal configuration for each dataset was selected based on the highest mean semantic similarity score achieved on the dev set. Finally, this single best configuration was evaluated once on the held-out test split to report our unbiased final metrics.

5 Results

5.1 Overall Performance

Table 1 summarizes the best development configuration for each dataset and its corresponding test performance. The strongest overall results were obtained on **Banking77**, which achieved a test semantic similarity of **0.656**. Performance was lower on the three news oriented datasets: **20 Newsgroups** reached **0.418**, **BBC News** reached **0.355**, and **AG News** reached **0.351**.

Dataset	Threshold	Top- k	α	Dev Sim	Test Sim	Test F1
BBC News	0.3	3	0.50	0.419	0.355	0.000
Banking77	0.3	10	0.50	0.667	0.656	0.338
20 Newsgroups	0.5	10	0.50	0.446	0.418	0.140
AG News	0.5	3	0.85	0.442	0.351	0.000

Table 1: Best development configuration and held-out test performance for each dataset.

These results indicate that the proposed method is capable of producing semantically meaningful cluster labels, but performance varies by dataset.

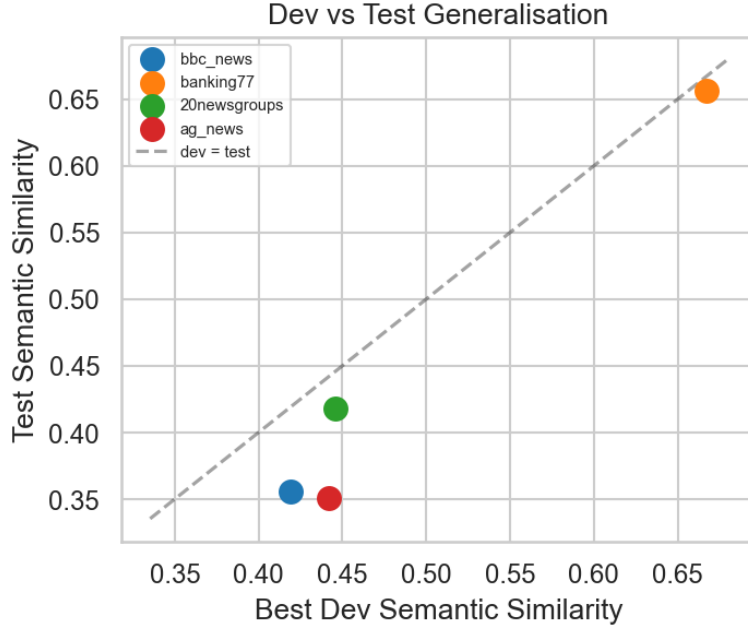


Figure 2: Development versus test semantic similarity for the best configuration selected on each dataset. Points close to the diagonal indicate stable generalization.

5.2 Development-to-Test Generalization

To assess generalization, we compared the best development score to the corresponding test score for each dataset. Banking77 showed the smallest drop, from 0.667 on development to 0.656 on test, a gap of only 0.011. 20 Newsgroups also generalized reasonably well, dropping from 0.446 to 0.418. In contrast, BBC News and AG News exhibited larger drops, from 0.419 to 0.355 and from 0.442 to 0.351 respectively.

Inspection of the generated labels (provided in Appendix A) shows that the dev-test gap is largely driven by differences in label granularity rather than complete topic drift. On Banking77, development and test labels are usually stable paraphrases of the same intent, whereas on BBC News and AG News the model tends to generate increasingly specific event or subtopic-level labels for broad category names such as *business*, *sport*, and *world*. 20 Newsgroups falls between these extremes: technical categories remain stable across splits, while political, religious, and medical categories shift toward narrower interpretive framings.

5.3 Hyperparameter Sensitivity

Figure 3 shows the effect of the three main hyperparameters on development-set semantic similarity, averaged over the remaining settings. Across datasets, no single hyperparameter configuration was optimal everywhere, but several consistent tendencies emerged.

First, lower or moderate similarity thresholds generally worked better than aggressive thresholds. The best threshold was 0.3 for BBC News and Banking77, and 0.5 for 20 Newsgroups and AG News. By contrast, the highest threshold setting of 0.7 didn’t produced a best run and often reduced performance. This trend is consistent with the role of the threshold in graph construction: a lower threshold produces a denser similarity graph, preserving more semantic links between documents within a cluster, whereas a high threshold yields a sparser graph that retains only the strongest pairwise similarities. For PageRank, an overly sparse graph is problematic because centrality then depends on only a small number of local connections, making the ranking less stable and less representative of the overall cluster structure. In contrast, a moderately dense graph allows PageRank to estimate document importance using a richer neighborhood structure, which is more likely to reflect the semantic core of the cluster rather than a small local pocket of highly similar documents.

Hyperparameter Sensitivity (dev set, averaged over other params)

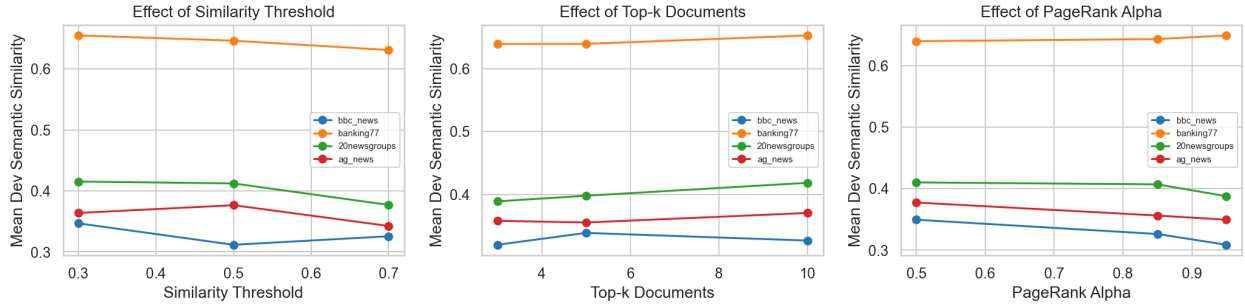


Figure 3: Average development semantic similarity as a function of the main hyperparameters: similarity threshold, number of representative documents (k), and PageRank damping factor α .

Second, the preferred number of representative documents depended on the dataset. Banking77 and 20 Newsgroups performed best with $k = 10$, whereas BBC News and AG News performed best with $k = 3$. This split is logically consistent with the granularity of the underlying clusters. Banking77 contains 77 fine-grained user intents, and 20 Newsgroups contains 20 relatively specific topical communities. In these datasets, adding more central documents usually reinforces the same underlying concept, giving the language model more evidence without substantially changing the semantic direction of the prompt. By contrast, BBC News and AG News contain only 5 and 4 broad categories, respectively. In these broader clusters, additional central documents are more likely to introduce different subtopics, events, or perspectives from within the same category. As a result, increasing k can dilute the prompt and pull the generated label away from the desired abstract category name toward a more fragmented or overly specific description. In this setting, a smaller value of k appears to act as a form of regularization, forcing the model to focus on the most dominant shared signal.

Third, the PageRank damping factor α showed weaker but still meaningful effects. A value of $\alpha = 0.5$ was selected as best for three of the four datasets, while AG News preferred $\alpha = 0.85$. In PageRank, α controls the balance between following graph edges and jumping uniformly to another node. Higher values place more weight on long random walks through the graph, making centrality more dependent on the recursive structure of document-to-document similarity. Lower values reduce that dependence and yield a flatter notion of centrality that is less sensitive to small structural irregularities. The fact that $\alpha = 0.5$ was most often preferred suggests that, in this task, document importance is captured sufficiently well by local connectivity and does not benefit from very strong recursive propagation. This is plausible because the graphs are already built from semantic similarity embeddings, so even local edges are informative. A higher α may occasionally help, as in AG News, when stronger propagation through the graph helps smooth over noise in a small number of broad classes, but overall the lower value appears more robust.

The hyperparameter sweep also revealed clear differences in stability across datasets. Banking77 was the most stable dataset, with semantic similarity varying only from 0.608 to 0.667 across all 27 configurations. BBC News and AG News were much more sensitive, each spanning a range of roughly 0.16 between their worst and best settings, while 20 Newsgroups fell in between. Figure 4 visualizes this pattern by showing the distribution of development scores across the full sweep. The narrow distribution for Banking77 indicates that many hyperparameter settings produce similar results, whereas the wider distributions for BBC News and AG News indicate that performance is much more dependent on how the graph is constructed and how many central documents are passed to the language model.

This stability pattern is also consistent with the structure of the datasets themselves. In semantically narrow and internally coherent clusters, such as the intent classes in Banking77, many documents are close to one another in embedding space, so moderate changes in threshold, damping, or k do not substantially alter which documents are selected as central. In broader and more heterogeneous clusters, the graph structure is more fragile: relatively small changes in the threshold can change connectivity, centrality rankings,

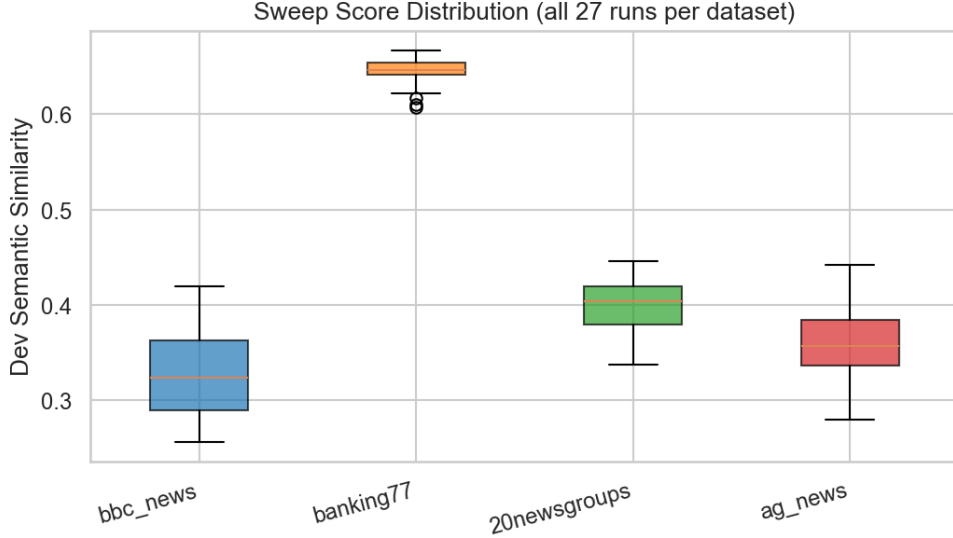


Figure 4: Distribution of development semantic similarity across all 27 hyperparameter configurations for each dataset. Narrower distributions indicate greater robustness to hyperparameter choice, while wider distributions indicate higher sensitivity.

and ultimately the evidence shown to the language model. This makes the final label more sensitive to hyperparameter choice.

5.4 Per-Cluster Analysis

Dataset-level averages conceal substantial variation in performance across individual clusters. This is visible in Figure 6, which shows that labeling quality is not uniform even within the same dataset.

In Banking77, many clusters achieve high semantic similarity, while a smaller number perform much worse. In 20 Newsgroups, strong and weak clusters coexist more evenly. In BBC News and AG News, the small number of broad topical categories leads to more limited but still noticeable variation. These patterns indicate that the difficulty of automatic cluster labeling depends not only on the dataset as a whole, but also on the internal structure of individual clusters.

To better understand this variation, we examined whether cluster-level labeling quality is associated with intrinsic cluster quality measures derived from the cluster-hypothesis literature in information retrieval. Specifically, we correlated per-cluster labeling scores on the test split with five structural and separability-based measures:

- **Jardine & van Rijsbergen’s (’71) Overlap Test:** we used a variation of this classic test [9], which evaluates the similarity between two relevant documents versus the similarity between a relevant and a non-relevant document. Specifically, it measures the overlap between these two similarity distributions. In our context, we adapted this to compare the similarity distribution of documents within the same cluster against the similarity distribution of documents belonging to different clusters. We decided to use this metric because it directly quantifies the fundamental assumption of the cluster hypothesis: that closely associated documents tend to be relevant to the same requests. By measuring this distributional overlap, we can assess whether a cluster’s internal cohesion is distinct enough from the rest of the corpus to provide a clear, unambiguous context for the LLM to generate an accurate label.
- **Voorhees’ (’85) Nearest-Neighbor Test:** we utilize a variation of this classic test [10]. The original test evaluates the cluster hypothesis by calculating the percentage of relevant documents among the k nearest neighbors of a known relevant document. In our cluster-labeling setting, we adapt this by first identifying the cluster medoid (the document with the highest mean cosine similarity to all other

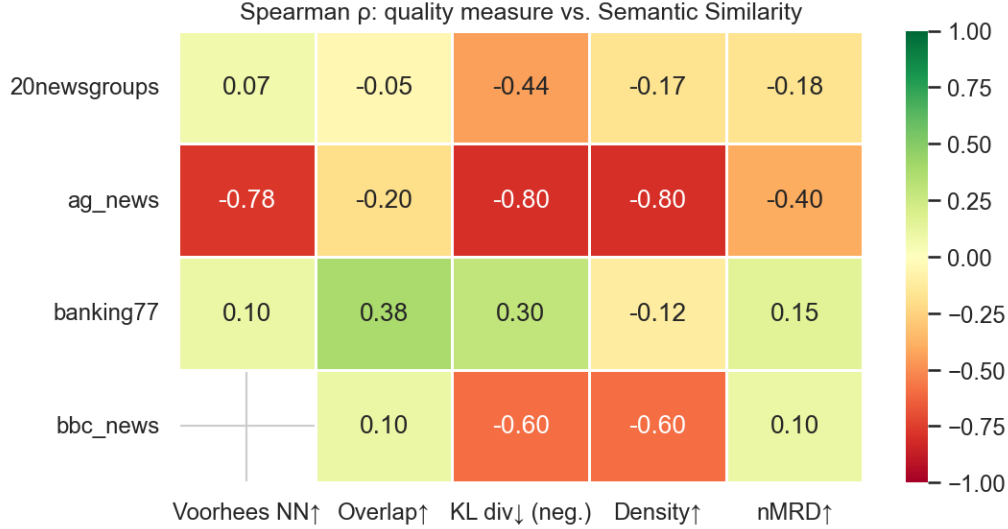


Figure 5: Correlation between cluster quality measures and per-cluster semantic similarity on the test split. Stronger positive values indicate that better cluster structure is associated with better labeling performance.

cluster members). Treating the medoid as the “relevant document” for the cluster, we then calculate the fraction of the medoid’s k global nearest neighbors that belong to the same cluster. This allows us to test whether the most representative document in a cluster is tightly surrounded by documents from the same class, indicating a strong semantic core.

- **Density-Based Cluster Hypothesis Test (El-Hamdouchi & Willett ’87):** we used this test, which calculates the ratio between the total number of terms in the cluster and the size of its unique vocabulary [11]. Higher values indicate that the documents within the cluster repeatedly use the same core set of terms. In the context of generative labeling, this test serves as a strong measure of lexical concentration and internal cohesion: a cluster with a high density score provides the LLM with a consistent and focused vocabulary, thereby reducing ambiguity and the risk of hallucination during label generation.
- **Voorhees’ Nearest-Neighbor Test using KL-Divergence (Raiber & Kurland ’12):** we used this test, which evaluates the neighborhood consistency among a set of documents by using the KL divergence between their unigram language models as the similarity measure [12]. In our context, we apply this to the documents within each cluster. Lower KL divergence indicates that the cluster members share highly similar term distributions (we negate this value in our correlation analysis so that a higher score reflects better cluster quality). We chose this language-model-based similarity measure because it naturally aligns with the generative mechanics of the LLMs used in our pipeline, capturing how consistently the documents “speak” the same language.
- **Alternative Cluster Hypothesis Test (Smucker & Allan ’09):** we evaluate the clusters using this test, which utilizes the Normalized Mean Reciprocal Distance (NMRD) [13]. This graph-based measure calculates how highly same-cluster documents rank in one another’s global similarity lists, capturing the local neighborhood consistency in the embedding space. Smucker and Allan proposed this metric to address the limitations of standard nearest-neighbor tests, arguing that a reciprocal rank-based distance provides a more accurate reflection of true clustering tendency. We chose this metric because our entire labeling pipeline is built upon Graph-Guided RAG, therefore, evaluating the clusters with an explicitly graph-based measure like NMRD perfectly aligns with our methodology for determining document centrality.

Figure 5 summarizes these relationships for semantic similarity.

The clearest pattern appears in Banking77. In this dataset, clusters that are better separated and more internally coherent tend to receive better labels. This is evident both qualitatively, from the strong concentration of high-scoring clusters in Figure 6, and quantitatively, from the correlation analysis. In particular, the strongest effects appear in the measures that most directly reflect separability from other clusters: overlap and negated KL-divergence. The strongest relationship was observed for the overlap measure, which correlated significantly with semantic similarity (Pearson $r = 0.392$, Spearman $\rho = 0.379$, both $p < 0.001$). KL-divergence also showed a significant positive association with semantic similarity after negation (Pearson $r = 0.286$, Spearman $\rho = 0.296$, both $p < 0.05$). These results suggest that, for fine-grained intent clusters, label generation works best when the cluster occupies a clear and well-separated region of the embedding space.

This finding is consistent with the qualitative behavior observed in Banking77. Highly coherent clusters such as `pending_cash_withdrawal` and `lost_or_stolen_card` were labeled very accurately, while broader or less tightly bounded clusters such as `supported_cards_and_currencies` were harder to summarize with a single short phrase. In other words, when the cluster itself has a strong semantic center, the graph-based retrieval stage is more likely to identify representative documents that support a clean and grounded label.

In contrast, the other datasets showed much weaker or inconclusive relationships between cluster quality and labeling performance. For 20 Newsgroups, none of the tested measures showed a statistically reliable correlation with semantic similarity or token-overlap F1, although KL-divergence exhibited a moderate negative trend with semantic similarity (Spearman $\rho = -0.441$, $p = 0.0519$). This suggests that per-cluster difficulty in 20 Newsgroups is not explained solely by geometric cohesion in embedding space: ideological diversity, argumentative structure, and topical breadth likely also play an important role. For BBC News and AG News, the number of clusters is too small ($n = 5$ and $n = 4$, respectively) to support strong statistical conclusions, so the correlation analysis should be interpreted cautiously.

Overall, these results indicate that cluster separability helps explain labeling success, but only in datasets where clusters are already semantically narrow and well defined. In broad topical datasets, cluster-level performance appears to depend not only on internal cohesion, but also on the level of abstraction required by the target label. This helps explain why some clusters receive low scores even when the generated label is thematically related to the documents: the system may correctly identify a dominant subtopic, yet still miss the broader category name expected by the benchmark.

5.5 Token Overlap F1

Token-overlap F1 was much lower than semantic similarity across all datasets, and especially low on BBC News and AG News, where test F1 dropped to 0.000. This does not necessarily imply total failure, as the generated labels were often semantically relevant but used different wording or a more specific phrasing than the reference label. For example, *Technology Industry News* is clearly related to *Sci/Tech* yet it shares no lexical tokens with the target label. This discrepancy highlights a classic vocabulary mismatch problem. In datasets with broad categories, we observe an "Abstraction Gap": the LLM, guided by the specific documents identified through PageRank, often generates a label that is a hyponym (a specific term) of the ground-truth hypernym (the broad category). Because our pipeline maximizes groundedness in the retrieved context, the model prioritizes factual precision over abstract benchmark names. Thus, the near-zero F1 scores primarily indicate a level-of-abstraction mismatch rather than a failure in thematic identification.

6 Conclusion and Future Work

This project proposed and evaluated a Graph-Guided Retrieval-Augmented Generation (RAG) framework for automatic cluster labeling. By constructing semantic similarity graphs and utilizing PageRank to select the most central documents, the pipeline successfully grounds LLM generation in representative context, minimizing the noise of full-cluster prompts.

The results validate the core intuition of our approach: selecting a small, central subset of documents produces highly accurate and meaningful labels for narrow, semantically cohesive clusters (as demonstrated by the strong performance on Banking77). However, the experiments also revealed a fundamental tradeoff regarding the level of abstraction. In broad, heterogeneous datasets (such as BBC News and AG News), graph centrality tends to isolate a dominant sub-topic rather than the overarching category theme. Consequently,

the LLM faithfully generates a label that is factually grounded in the retrieved documents, but often too specific relative to the desired category label.

To address these limitations and expand upon the current framework, we identify three primary directions for future work:

- **Diversity-Aware Selection:** The current methodology relies strictly on centrality, which can lead to over-specific labels in broad clusters if the top-ranked documents all represent the same local sub-theme. A natural extension would be to balance centrality with diversity (e.g., using Maximal Marginal Relevance) to ensure the LLM prompt covers multiple distinct regions of the cluster.
- **Baseline Comparison:** While the current methodology explored hyperparameter sensitivity across similarity thresholds and PageRank damping factors, a natural direction for future work is the inclusion of a random k-selection baseline. Comparing labels generated from PageRank-selected documents against those generated from randomly sampled documents of equivalent size would empirically isolate the contribution of the graph-filtering mechanism from the generative capabilities of the underlying LLM, providing a clearer picture of where the performance gains actually originate.
- **Human Evaluation:** As demonstrated by the near-zero F1 scores on several datasets, strict lexical overlap metrics heavily penalize valid paraphrases and descriptive labels. While semantic similarity mitigates this, it does not perfectly capture the nuance of abstraction levels. Future work should incorporate human evaluation to assess label usefulness, groundedness, and specificity.

Ultimately, graph-guided RAG proves to be a highly effective strategy for cluster labeling. By balancing centrality with broader coverage in future iterations, the framework can be further refined to produce labels that are both strictly grounded and appropriately general.

A Generated Label Comparison

This appendix reports the ground-truth cluster names together with the labels generated on the development and test splits under the best configuration selected for each dataset. These tables support the analysis in the Development-to-Test Generalization subsection by showing whether the model remains at the same level of semantic abstraction across splits or shifts toward a narrower event or subtopic-level description.

A.1 BBC News

Table 2: Ground-truth labels and generated labels on the development and test splits for BBC News.
Best configuration: threshold = 0.3, top- k = 3, α = 0.5.

Ground Truth	Dev Generated Label	Test Generated Label
business	Business and Economic Trends	US Corporate Consolidation
entertainment	Celebrity and Entertainment News	Celebrity and Film Awards
politics	UK Labour Party Election	UK Conservative Policy
sport	English Rugby Team Dynamics	Six Nations Rugby
tech	Digital Lifestyle Trends	Mobile Technology Adoption

A.2 Banking77

Table 3: Ground-truth labels and generated labels on the development and test splits for Banking77.
Best configuration: threshold = 0.3, top- k = 10, α = 0.5.

Ground Truth	Dev Generated Label	Test Generated Label
Refund_not_showing_up	Missing Refund Issues	Missing Refund Issues
activate_my_card	Card Activation Instructions	Card Activation Instructions
age_limit	Account Age Requirements	Account Age Requirements
apple_pay_or_google_pay	Apple Pay Top Up	Apple Pay Top Up Issues
atm_support	ATM Location Inquiry	ATM Withdrawal Locations
automatic_top_up	Auto Top-up Options	Auto Top-up Options
balance_not_updated_after_bank_transfer	Missing Account Transfers	Delayed Transaction Issues
balance_not_updated_after_cheque_or_cash_deposit	Delayed Deposits Issue	Missing Deposits Issue
beneficiary_not_allowed	Transfer Issues with Beneficiaries	Transfer Failure Issues
cancel_transfer	Transfer Cancellation Requests	Canceling Incorrect Transfers
card_about_to_expire	Card Renewal Process	Credit Card Renewal
card_acceptance	Card Acceptance Locations	Card Acceptance Locations
card_arrival	Card Delivery Issues	Card Delivery Issues
card_delivery_estimate	Card Delivery Timing	Delivery Timeframe Inquiry
card_linking	Linking a New Card	Linking Existing Cards
card_not_working	Card Functionality Issues	Card Functionality Issues
card_payment_fee_charged	Card Payment Fees	Card Payment Fees
card_payment_not_recognised	Unrecognized Card Payments	Unauthorised Card Transactions
card_payment_wrong_exchange_rate	Exchange Rate Disputes	Incorrect Exchange Rates
card_swallowed	ATM Card Retrieval Issues	ATM Card Retrieval Issues

Ground Truth	Dev Generated Label	Test Generated Label
cash_withdrawal_charge	Cash Withdrawal Fees	Cash Withdrawal Fees
cash_withdrawal_not_recognised	Unauthorized Cash Withdrawals	Unrecognized Cash Withdrawals
change_pin	PIN Change Instructions	PIN Change Assistance
compromised_card	Card Identity Theft	Unauthorized Card Use
contactless_not_working	Contactless Payment Issues	Contactless Payment Issues
country_support	International Card Availability	Geographic Service Availability
declined_card_payment	Payment Decline Queries	Card Payment Declines
declined_cash_withdrawal	ATM Withdrawal Issues	ATM Withdrawal Issues
declined_transfer	Bank Transfer Declines	Bank Transfer Issues
direct_debit_payment_not_recognised	Unrecognized Direct Debits	Unauthorized Direct Debits
disposable_card_limits	Disposable Card Limits	Disposable Card Restrictions
edit_personal_details	Account Information Update	Account Detail Changes
exchange_charge	Currency Exchange Fees	Currency Exchange Fees
exchange_rate	Exchange Rate Calculation	Foreign Exchange Rates
exchange_via_app	Currency Exchange	Currency Exchange
extra_charge_on_statement	Extra Statement Fees	Unexplained Bank Charges
failed_transfer	Transfer Failure Issues	Transfer Failure Issues
fiat_currency_support	Supported Fiat Currencies	Currency Support Options
get_disposable_virtual_card	Disposable Virtual Cards	Disposable Virtual Cards
get_physical_card	PIN Retrieval Issues	Card PIN Inquiry
getting_spare_card	Requesting Additional Cards	Requesting Additional Cards
getting_virtual_card	Virtual Card Acquisition	Virtual Card Acquisition
lost_or_stolen_card	Lost or Stolen Cards	Lost or Stolen Cards
lost_or_stolen_phone	Lost or Stolen Phones	Phone Loss and Security
order_physical_card	Physical Card Acquisition	Physical Card Requests
passcode_forgotten	Passcode Reset Issues	Passcode Recovery Issues
pending_card_payment	Pending Card Payments	Pending Card Payments
pending_cash_withdrawal	Pending Cash Withdrawals	Pending Cash Withdrawals
pending_top_up	Pending Mobile Top-Ups	Pending Top-Up Issues
pending_transfer	Transfer Status Inquiry	Transfer Status Issues
pin_blocked	PIN Unblock Issues	PIN Unblock Issues
receiving_money	Currency Exchange Inquiry	Currency Salary Options
request_refund	Refund Requests	Refund Request Process
reverted_card_payment?	Card Payment Issues	Card Payment Issues
supported_cards_and_currencies	Payment Options and Support	Payment Options Inquiry
terminate_account	Account Deletion Requests	Account Closure Requests
top_up_by_bank_transfer_charge	Transfer Fees and Charges	Transfer Fees and Charges
top_up_by_card_charge	Card Top-Up Fees	Top Up Charges
top_up_by_cash_or_cheque	Cheque Account Top Up	Account Top-Up Methods
top_up_failed	Failed Mobile Top-Ups	Failed Top-Up Issues
top_up_limits	Top-up Limit Inquiry	Top-up Limit Inquiry
top_up_reverted	Top-up Reversal Issues	Top-Up Reversal Issues
topping_up_by_card	Missing Top-Up Funds	Card Top Up Issues
transaction_charged_twice	Duplicate Transaction Charges	Duplicate Charges
transfer_fee_charged	Transfer Fee Complaints	Money Transfer Fees
transfer_into_account	Bank Account Transfers	Account Transfer Process
transfer_not_received_by_recipient	Unsuccessful Money Transfers	Unsuccessful Money Transfers

Ground Truth	Dev Generated Label	Test Generated Label
transfer_timing	Transfer Processing Time	Transfer Wait Times
unable_to_verify_identity	Identity Verification Issues	Identity Verification Issues
verify_my_identity	Identity Verification Requirements	Identity Verification Process
verify_source_of_funds	Funding Source Inquiry	Source of Funds Verification
verify_top_up	Top-up Card Verification	Top-up Verification Process
virtual_card_not_working	Virtual Card Issues	Virtual Card Issues
visa_or_mastercard	Credit Card Choice	Credit Card Options
why_verify_identity	Identity Verification Requirements	Identity Verification Requirements
wrong_amount_of_cash_received	ATM Cash Disputes	ATM Withdrawal Issues
wrong_exchange_rate_for_cash_withdrawal	Exchange Rate Complaints	Incorrect Exchange Rates

A.3 20 Newsgroups

Table 4: Ground-truth labels and generated labels on the development and test splits for 20 Newsgroups.
Best configuration: threshold = 0.5, top- k = 10, α = 0.5.

Ground Truth	Dev Generated Label	Test Generated Label
alt.atheism	Atheism and Critique	Atheism and Free Speech
comp.graphics	Computer Graphics Software	Computer Graphics Software
comp.os.ms-windows.misc	Microsoft Windows Discussion	Windows Software Development
comp.sys.ibm.pc.hardware	486 PC Hardware Discussions	Computer Hardware Reviews
comp.sys.mac.hardware	Macintosh Hardware Discussion	Macintosh Hardware Discussions
comp.windows.x	X Window System Programming	X Window System Programming
misc.forsale	Used Electronics Marketplace	Computer Equipment Sales
rec.autos	Used Car Discussions	Automotive Discussion Forum
rec.motorcycles	Motorcycle Enthusiasts	Motorcycle Enthusiasts' Forum
rec.sport.baseball	Baseball Statistical Analysis	Baseball Analysis and Discussion
rec.sport.hockey	NHL Hockey Discussion	NHL Hockey Fan Discussions
sci.crypt	Clipper Chip Controversy	Clipper Chip Opposition
sci.electronics	Electronics and Hacking	Computer Hardware Discussions
sci.med	Critique of Conventional Medicine	Alternative Health Perspectives
sci.space	Criticism of NASA Funding	Space and Astronomy Issues
soc.religion.christian	Christianity and Critique	Christian Biblical Interpretation
talk.politics.guns	Gun Rights Debate	Gun Rights Debate
talk.politics.mideast	Israel-Palestine Conflict	Israel-Palestine Debate
talk.politics.misc	Anti-Government Sentiment	Anti-Government and Homophobia
talk.religion.misc	Christianity Critique and Debate	Cult and Christianity Critique

A.4 AG News

Table 5: Ground-truth labels and generated labels on the development and test splits for AG News.
Best configuration: threshold = 0.5, top- k = 3, α = 0.85.

Ground Truth	Dev Generated Label	Test Generated Label
Business	Corporate Earnings Reports	US Corporate Earnings
Sci/Tech	Tech Industry Developments	Technology Industry News
Sports	US Sports News	International Cricket News
World	Middle East Conflict	US Military Conflicts

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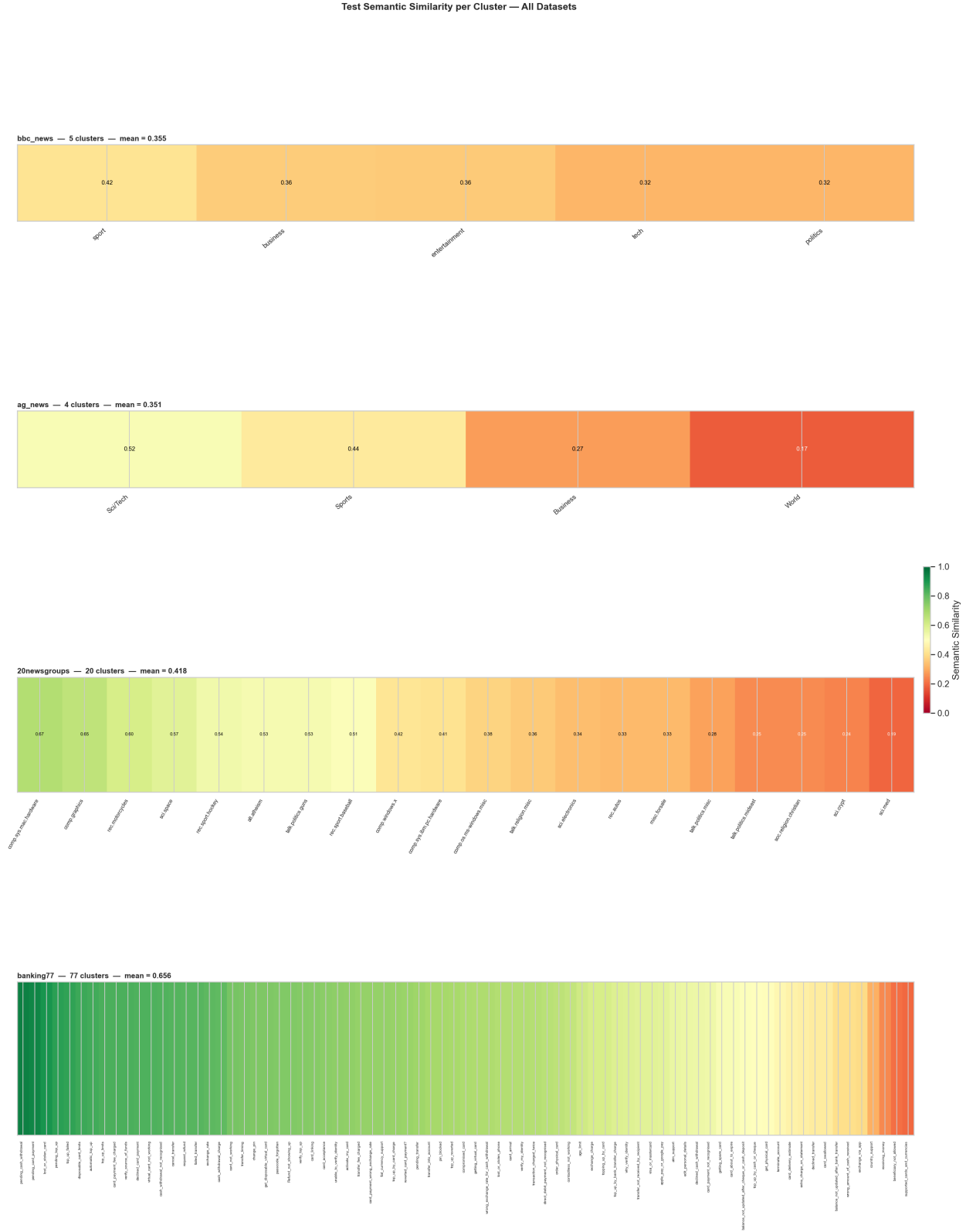


Figure 6: Per-cluster test semantic similarity across all datasets. The figure highlights the large within-dataset variation that is hidden by dataset-level averages: some clusters are labeled very accurately, while others remain difficult even within the same dataset.